

**FRESHROUTE - A SMART FRUIT SUPPLY CHAIN SYSTEM TO REDUCE
WASTE AND ELIMINATE MIDDLEMEN IN SRI LANKA**

Draft Thesis Report

Gallearachchi S. C.

(IT22230874)

B.Sc. (Hons) Degree in Information Technology Specializing in Software Engineering

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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DECLARATION

I declare that this dissertation is my own work and does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where such acknowledgement is made in the text.

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Candidate: Gallearachchi S C

Student ID: IT22230874

Signature: _____ Date: _____

The above candidate has carried out research for the bachelor's degree dissertation under my supervision.

Supervisor: _____ Date: _____

Co-supervisor: _____ Date: _____

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I would like to express my sincere gratitude to my supervisor and co-supervisor for the continuous guidance, comments, and encouragement provided throughout this research journey. Their academic support helped shape the work from a broad group project idea into a more focused individual study on the logistics side of the FreshRoute system. I am equally thankful for the opportunity to develop this work within a wider multidisciplinary team, because the forecasting, trading, and quality-assurance perspectives of the overall project helped me understand how logistics must operate as part of an end-to-end fruit supply chain rather than as an isolated routing tool.

I also wish to acknowledge the individuals who shared practical insights about fruit handling, transport delays, and the realities faced by smallholder farmers and market actors in Sri Lanka. These discussions made it clear that reducing spoilage requires more than technology alone; it requires a solution that fits local road conditions, seasonal variations, financial limitations, and human coordination constraints. Finally, I would like to thank my family and friends for their patience and support, and the Faculty of Computing for providing the environment and resources needed to complete this thesis.

ABSTRACT

Sri Lanka's fruit industry continues to suffer from post-harvest losses caused by delayed transport, poor vehicle selection, weak cold-chain visibility, and fragmented coordination among farmers, buyers, transporters, and market intermediaries. Although digital trading platforms and advisory systems have emerged, logistics decisions are still frequently made manually, causing highly perishable fruit batches to be moved in unsuitable vehicles or along inefficient routes. This dissertation presents a logistics-focused thesis for the FreshRoute project, a smart fruit supply-chain platform intended to reduce spoilage and improve the movement of fresh produce from farms to buyers. The study focuses on the individual component responsible for spoilage-aware transport allocation, route optimisation, and in-transit monitoring. The research draws on Sri Lankan post-harvest literature, project planning documents, and a detailed analysis of the logistics problem faced by smallholder-oriented supply chains. A design-oriented methodology is used to identify operational pain points, define requirements, design the system architecture, and evaluate the prototype through scenario-based logistics testing. The proposed module combines rule-based vehicle selection, route planning, telemetry-driven monitoring, and alert management so that different fruit batches can be matched with suitable transport conditions according to perishability, handling requirements, and delivery urgency. The document also examines commercialization potential, adoption barriers, and operational governance considerations. Scenario-based evaluation suggests that the FreshRoute logistics workflow can reduce travel delays, improve delivery visibility, and lower spoilage risk when compared with informal manual planning. The study concludes that the logistics module represents a practical and scalable contribution to the wider FreshRoute platform, while also highlighting the need for field pilots, stronger data governance, and broader farmer onboarding for real-world deployment.

Keywords: post-harvest loss, fruit logistics, vehicle allocation, route optimisation, cold-chain monitoring, Sri Lanka

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

API - Application Programming Interface

B2B - Business-to-Business

B2C - Business-to-Consumer

ETA - Estimated Time of Arrival

GPS - Global Positioning System

IoT - Internet of Things

NFR - Non-functional Requirement

PHL - Post-harvest Loss

UI/UX - User Interface / User Experience

VRP - Vehicle Routing Problem

WBS - Work Breakdown Structure

1 INTRODUCTION

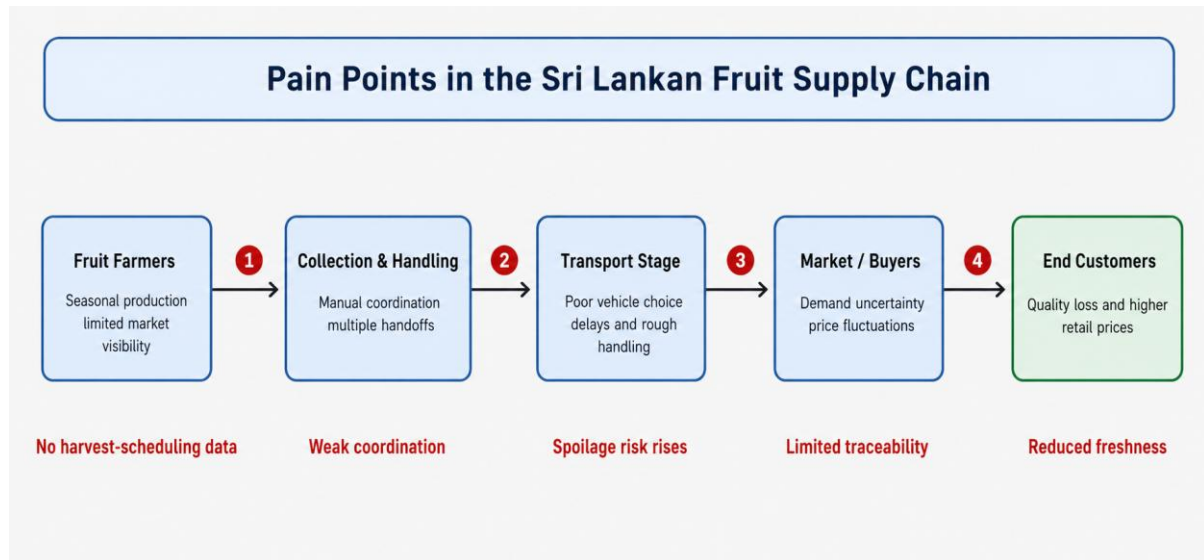


Figure 1.1: Pain points in the Sri Lankan fruit supply chain targeted by FreshRoute.

Table 1.1: Summary of selected literature relevant to the FreshRoute logistics problem.

Study	Main focus	Most relevant insight for the logistics module
Rajapaksha et al. [1]	National review of fruit and vegetable losses	Handling, transport, and weak cold-chain integration remain major drivers of loss.
Kamalakkannan et al. [2]	Banana post-harvest life-cycle assessment	Transport and downstream handling materially affect quality and waste.
Herath et al. [5]	Improved mango post-harvest practices	Better transport-related handling can reduce damage and extend shelf life.
Samarasinghe et al. [8]	Traceability feasibility in Sri Lankan supply chains	Low traceability weakens accountability and operational learning.
Osvald and Stirn [13]	Perishable vehicle routing	Routing should consider perishability rather than distance alone.

Table 1.2: Research gap comparison matrix.

Feature / gap	Training or packaging interventions	Marketplace-only concepts	FreshRoute logistics module
Perishability-aware vehicle assignment	No	No	Yes

Route planning tied to confirmed orders	No	Partial	Yes
Telemetry and alert visibility	Partial	No	Yes
Digital audit trail for delivery events	No	Partial	Yes
Smallholder-oriented coordination workflow	Partial	Partial	Yes

Table 1.3: Alignment between research questions and objectives.

Research question	Related objective
How can the module reduce fruit spoilage through logistics?	Design and explain a spoilage-aware transport workflow.
Which factors should influence vehicle selection?	Define decision rules for vehicle suitability.
How can routing support freshness preservation?	Present and evaluate a practical routing workflow.
How can telemetry improve accountability?	Design monitoring and alert mechanisms.
What is needed for adoption and deployment?	Discuss commercialization, governance, and future work.

1.1 Background and context

Sri Lanka's fruit supply chain is economically significant but operationally fragile. Fruits such as banana, mango, papaya, pineapple, and avocado support household income, rural employment, and domestic food distribution, yet a large proportion of harvested produce still loses market value before reaching consumers. The problem is especially severe for smallholder farmers who operate with thin margins, limited access to cold storage, and weak bargaining power. In many parts of the country, produce is harvested on the basis of rough market expectations and then moved through collectors, transporters, wholesalers, and retailers using loosely coordinated arrangements. While these arrangements may move produce quickly under favourable conditions, they are highly vulnerable to delay, overloading, route inefficiency, and inconsistent quality handling.

Transport-related spoilage is not simply a matter of distance. Fresh fruit can deteriorate because the wrong vehicle type is used, because fragile produce is mixed with incompatible loads, because protective packaging is insufficient, because ambient temperature is too high, or because long waiting times convert a short route into a freshness failure. Smallholders frequently depend on the next available truck rather than on a planned service that reflects perishability. As a result, decisions that should be based on crop sensitivity and delivery urgency are instead based on availability, habit, or cost alone. This disconnect is one of the main reasons why post-harvest losses remain persistent in the Sri Lankan context.

FreshRoute was conceived as a digital response to this broader supply-chain problem. The overall group project proposed an integrated platform that combines demand forecasting, direct trading, quality verification, and logistics coordination. Within that platform, the

present individual component focuses on the logistics layer: how to allocate suitable vehicles, plan efficient trips, monitor conditions during transit, and respond to disruptions before a batch loses value. The logistics component is central because every order ultimately becomes a physical movement problem. Even if forecasting is accurate and trading is transparent, the promised value collapses if the fruit arrives late, warm, bruised, or untraceable.

1.2 Background literature and literature survey

Research on post-harvest loss in Sri Lanka consistently shows that weaknesses after harvesting remain a major barrier to food security and farmer income. Reviews of the fruit and vegetable sector report that losses occur due to poor harvesting practices, weak grading, poor packaging, unsuitable storage, and transport delays. These studies emphasize that the problem is systemic rather than isolated: a fruit batch can be damaged slightly at harvest, stressed again during loading, overheated in transit, and finally discounted because it reaches the market in inconsistent condition. Such multi-stage deterioration means that logistics decisions must be made with knowledge of perishability rather than with a pure cost-minimization mindset.

Commodity-specific studies reinforce this argument. Work on banana supply chains in Sri Lanka indicates that transportation and downstream handling significantly contribute to loss and quality degradation. Studies on mango have similarly shown that improved post-harvest handling, better packaging, and more careful logistics can increase shelf life and reduce visible damage. These studies are important for the FreshRoute logistics module because they show that transport is not a passive link between two locations; it is a high-risk stage where mechanical stress, heat exposure, and travel delay materially affect final quality.

Another relevant strand of literature addresses the structure of the value chain itself. Horticultural studies from Sri Lanka describe supply channels in which farmers often sell through collectors and intermediary actors who control access to urban markets. This structure may simplify immediate sale for the farmer, but it can also reduce price transparency and weaken accountability for transport-related damage. When the same batch changes hands multiple times, it becomes difficult to determine where deterioration occurred and who should bear responsibility for the loss. A digital system that combines logistics records with event histories therefore has potential not only to improve operational efficiency but also to improve trust.

Traceability research adds a further dimension. Studies on fruit and vegetable traceability in Sri Lanka suggest that record-keeping practices remain uneven and that the feasibility of end-to-end traceability is limited by fragmented data collection, inconsistent product identification, and low technology adoption in some parts of the chain. This is highly relevant to the logistics component because a transport decision should ideally leave a digital trail: which vehicle carried the fruit, when it left, what the route was, whether temperatures stayed within the target range, whether delays occurred, and what condition was recorded at

delivery. Without this evidence, process improvement becomes difficult and complaint handling remains subjective.

International logistics research provides useful technical foundations for such a system. Vehicle Routing Problem literature and its perishable extensions show that routing decisions can be adapted to include time windows, heterogeneous fleets, and freshness decay. Studies on smart logistics additionally show the value of integrating sensor data and live transport status into dispatch decisions. Yet many of these models assume mature cold-chain infrastructure and relatively formalised supply-chain networks. Sri Lanka, especially at smallholder level, requires a more context-sensitive approach that can function with limited refrigerated capacity, mixed road conditions, and the practical need for human-readable rules.

1.3 Existing systems and practical context

Several initiatives already address parts of the fruit supply-chain problem, but few combine digital trading and logistics in a unified operational workflow. Traditional market arrangements remain heavily dependent on collectors, middlemen, and ad hoc transport hiring. In parallel, some online marketplaces allow products to be listed digitally, but the burden of arranging collection and delivery is often left to the users themselves. In such systems, the information layer is digitized while the physical movement layer remains informal and weakly governed.

Public-sector interventions such as packaging programmes and post-harvest training have improved awareness of crate use and safer handling. These are valuable contributions, yet they do not automatically create route visibility, dynamic vehicle assignment, or real-time exception response. Corporate cold-chain solutions exist as well, but they are usually designed for large, organized supply networks rather than for distributed smallholder ecosystems. This leaves a gap for a solution that supports affordable, modular, and operationally explainable logistics for a wider range of users.

FreshRoute therefore positions itself not merely as a tracking application but as a coordinating system. Its logistics component is expected to receive confirmed orders, understand the characteristics of the fruit to be moved, decide which vehicle type is required, generate a practical route, issue notifications, and preserve a record of what happened during the journey. In that sense, the module sits between digital trade and physical distribution.

1.4 Research gap

The first research gap lies in the separation between market-facing systems and transport execution. Many studies document that spoilage occurs during transport, yet existing solutions often stop at advisory recommendations, packaging training, or online produce visibility. Few systems make transport decisions on the basis of perishability. In practice, this means that the operational stage where quality can deteriorate fastest remains weakly supported by digital tools.

The second gap concerns real-time decision support. Research has quantified post-harvest losses and identified reasons such as temperature exposure, poor handling, and delays; however, the same literature rarely shows how a working system would detect those conditions during a live trip and react to them. A suitable logistics module should not only plan the trip at the start but also monitor its condition, warn users when a cold-chain deviation appears, and create an auditable event history.

The third gap concerns contextual suitability for Sri Lanka. Many advanced routing models assume abundant reefer fleets, standardized packaging, or highly digitized supply networks. Smallholder-oriented chains in Sri Lanka are different. They need a lighter-weight architecture that can still reason about freshness, route practicality, and communication constraints without assuming premium infrastructure at every step. The FreshRoute logistics module addresses this gap by emphasizing rule-based operational suitability, modular telemetry, and actor-specific dashboards that can be scaled gradually.

1.5 Research problem

The core research problem addressed in this thesis is the absence of a practical, spoilage-aware logistics mechanism for Sri Lankan fruit distribution that can allocate appropriate vehicles, plan efficient routes, and maintain visibility over the delivery process. At present, transport decisions are often made manually, based on whatever truck is available or cheapest at the moment. Such decisions do not reliably consider the perishability of the fruit, the length of the route, prevailing weather conditions, or the consequences of delay.

Because of this, highly perishable fruit may be transported in open or unsuitable vehicles, mixed with other loads that accelerate deterioration, or delayed without anyone receiving timely information. The research problem therefore extends beyond routing alone. It includes the need to coordinate multiple actors, preserve transport evidence, and support alert-driven intervention when delivery conditions become unsafe.

1.6 Research questions

The thesis is guided by five research questions. First, how can a digital logistics module reduce post-harvest fruit spoilage in Sri Lanka through better transport allocation and route planning? Second, which operational factors should influence vehicle selection for perishable fruit deliveries? Third, how can route optimisation be adapted to balance travel efficiency with freshness preservation and service reliability? Fourth, how can telemetry and alerts improve visibility and accountability during transport? Fifth, what business and deployment conditions are required for the logistics module to be adopted in real farming and buyer networks?

1.7 Research objectives

The main objective of this study is to design and document an individual logistics component for the FreshRoute project that reduces transport-related fruit spoilage by combining vehicle

allocation, route optimisation, and condition monitoring. To achieve this objective, the thesis pursues several supporting goals: to analyse the logistics-related pain points in the Sri Lankan fruit supply chain, to derive functional and non-functional requirements for the module, to present an architecture and workflow for spoilage-aware transport, to evaluate the module through scenario-based testing, and to discuss commercialization and deployment considerations.

A further objective is to separate the logistics contribution from the broader group system while still showing how it integrates with forecasting, trading, and quality-management functions. This is important because the thesis is an individual document and must therefore show the technical and research contribution of the logistics module in its own right.

1.8 Scope and delimitations

This dissertation focuses specifically on the logistics side of the FreshRoute concept. It examines how transport allocation, routing, monitoring, and related decision workflows can be designed for perishable fruit movement. The study refers to the wider FreshRoute platform only where necessary to explain integration points such as order confirmation, notifications, quality review, and payment status. It does not attempt to fully document or evaluate the forecasting, marketplace, or image-verification components as individual research contributions.

The evaluation reported in this study is scenario-based rather than field-based. That means the conclusions are framed around prototype logic, design consistency, and simulated operational benefits rather than around long-term performance in a live commercial deployment. This boundary is deliberate and appropriate for a final-year thesis, but it also means that future field trials are essential before claims about national-scale spoilage reduction can be validated.

1.9 Chapter summary

Chapter 1 introduced the logistics problem in the Sri Lankan fruit supply chain, reviewed relevant literature, identified the research gap, and stated the objectives and boundaries of the study. The next chapter explains the methodology used to build the logistics-focused thesis, gather requirements, design the module, and evaluate the proposed prototype in a structured manner.

2 METHODOLOGY

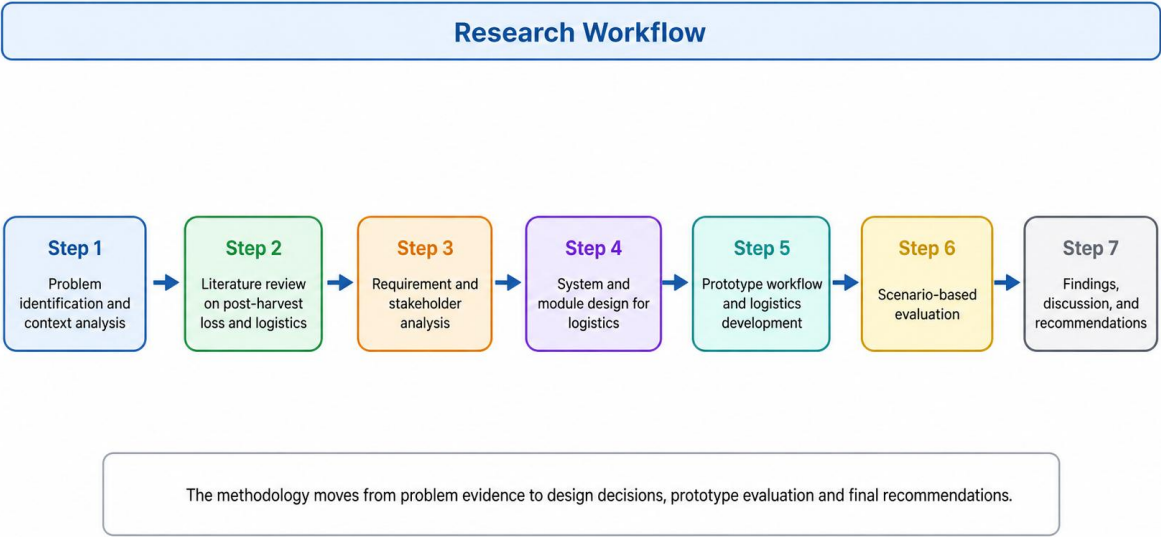


Figure 2.1: Research workflow for the logistics-focused FreshRoute thesis.

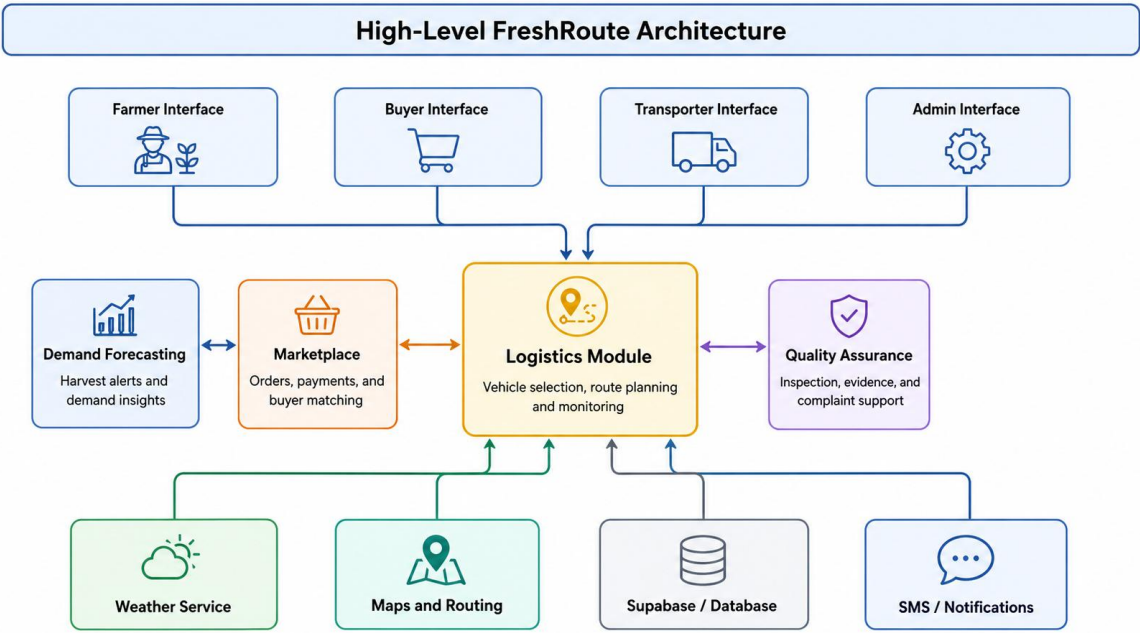


Figure 2.2: High-level FreshRoute architecture with emphasis on the logistics module.

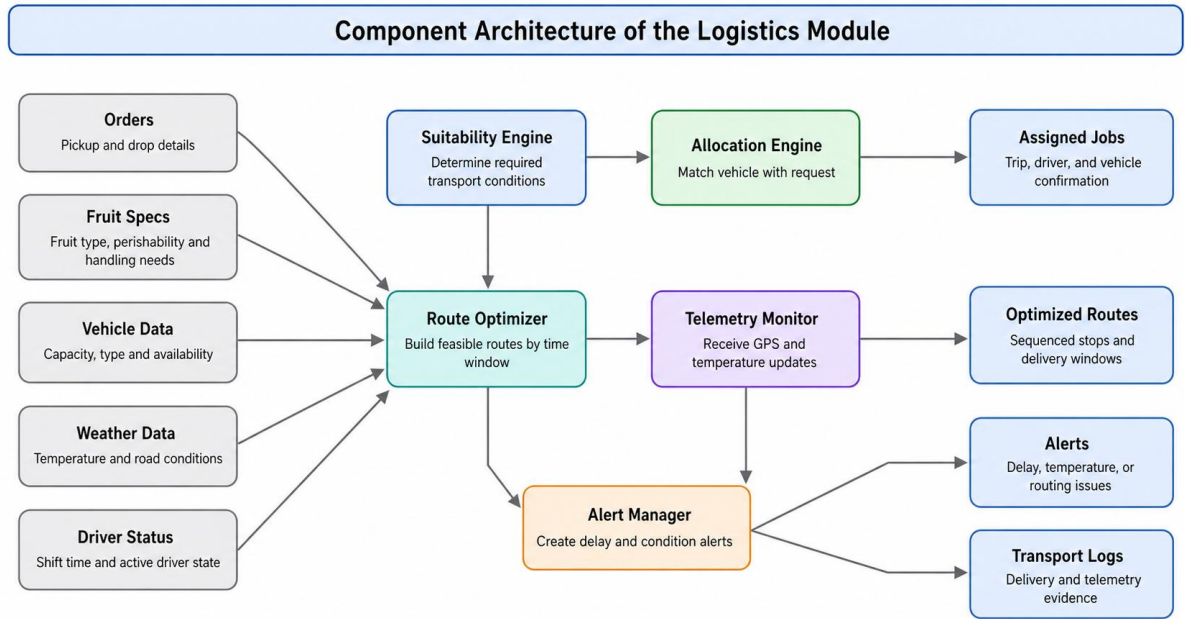


Figure 2.3: Component architecture of the spoilage-aware logistics module.

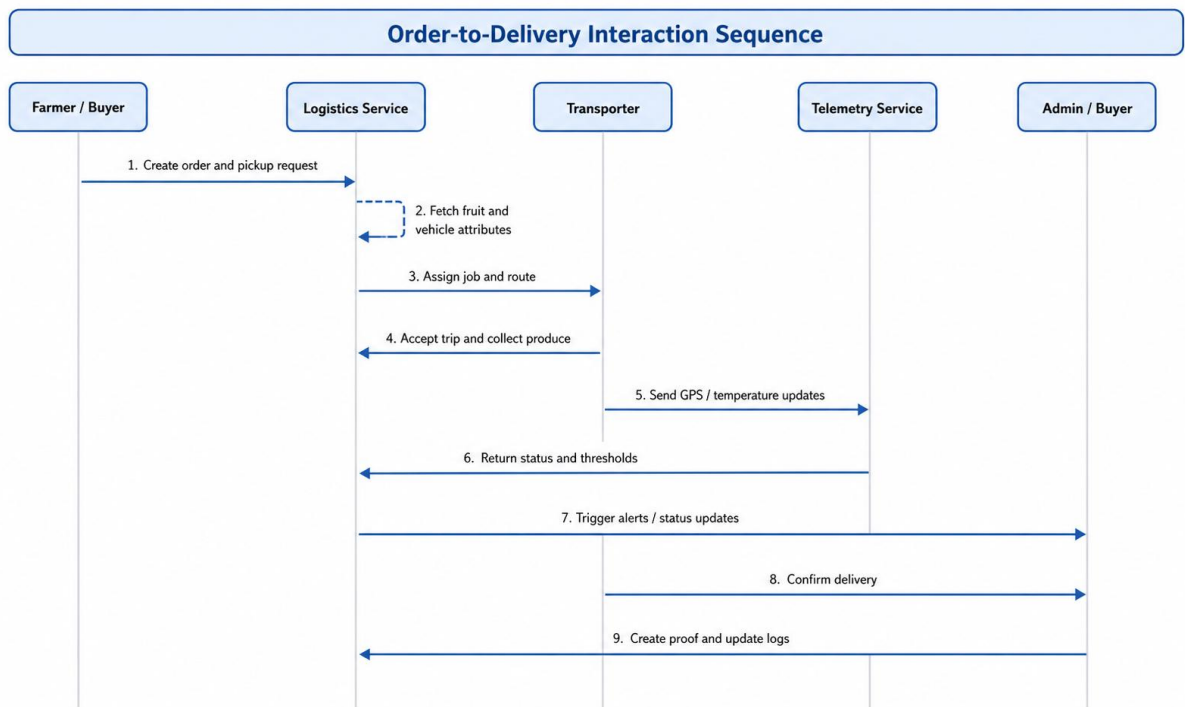


Figure 2.4: Order-to-delivery interaction sequence for the logistics workflow.

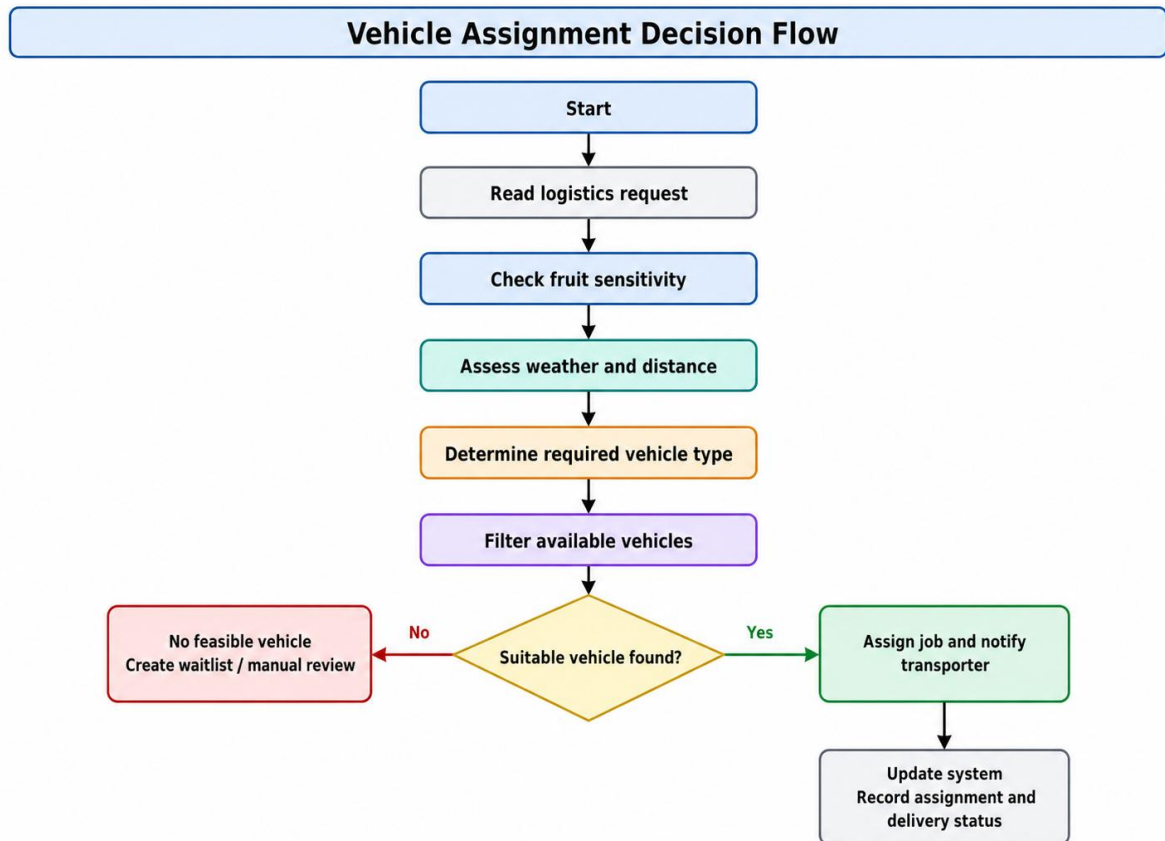


Figure 2.5: Vehicle assignment decision flow.

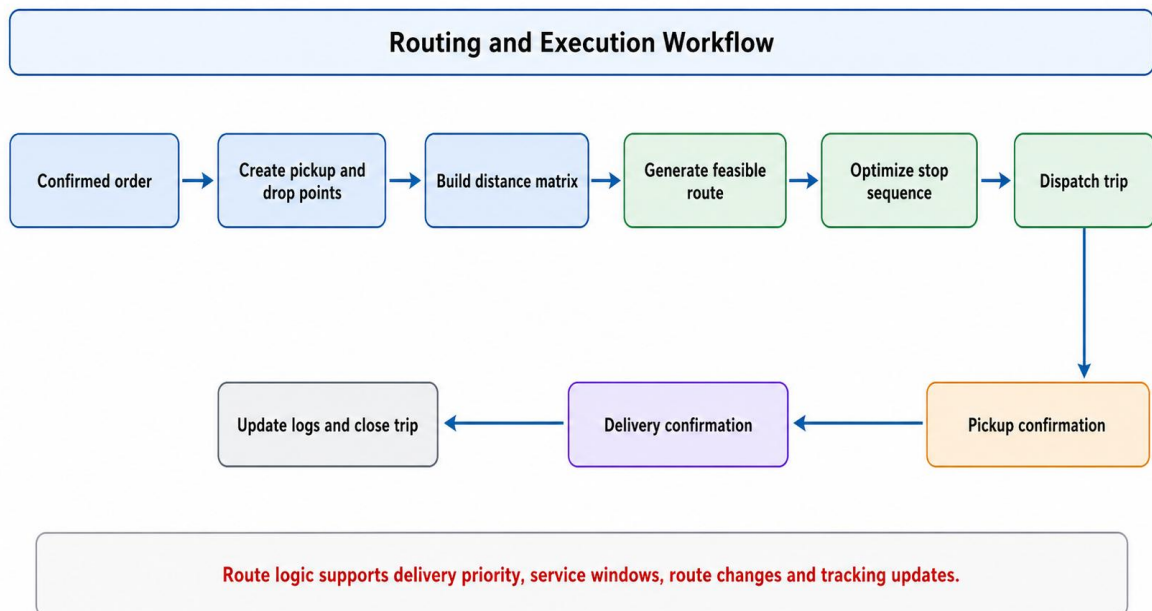


Figure 2.6: Routing and execution workflow.

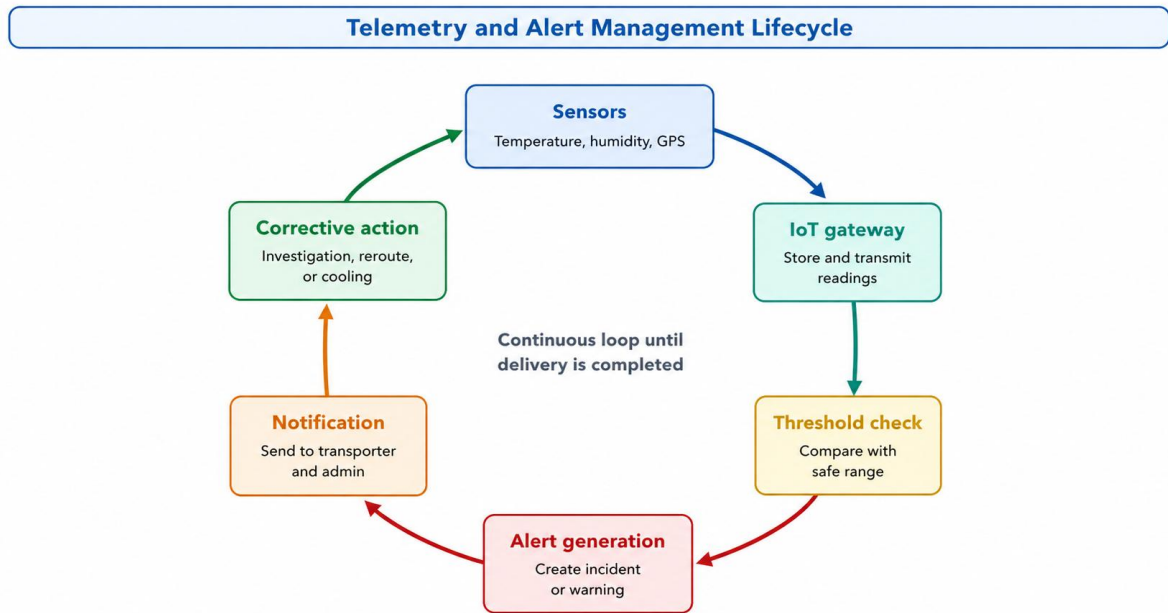


Figure 2.7: Telemetry and alert management lifecycle.

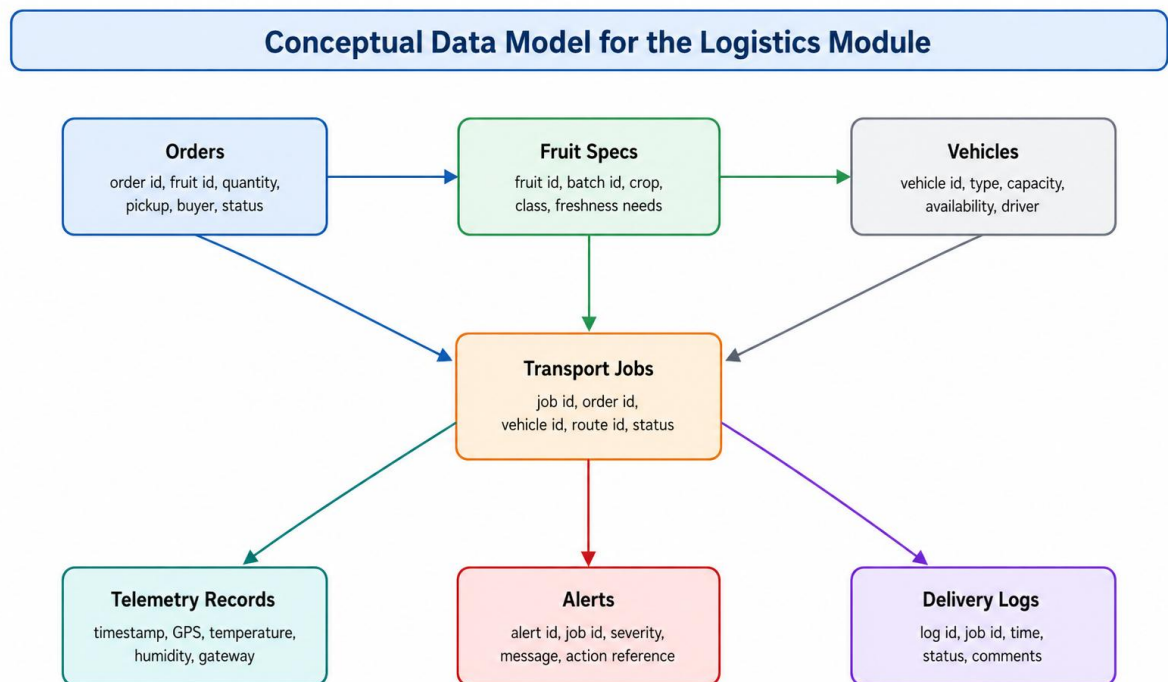


Figure 2.8: Conceptual data model for the logistics module.

Table 2.1: Stakeholder roles and logistics expectations.

Stakeholder	Main logistics expectation
Farmer	Simple pickup request, timely collection, and reduced spoilage at farm gate.

Buyer	Reliable ETA, quality-preserving transport, and delivery visibility.
Transporter	Clear work order, route guidance, and manageable exception handling.
Administrator	Fleet oversight, alert review, and auditable service history.

Table 2.2: Functional requirements of the logistics module.

Requirement ID	Requirement
FR-01	Create logistics requests from confirmed fruit orders.
FR-02	Select suitable vehicle classes according to perishability and route context.
FR-03	Generate feasible pickup and delivery routes.
FR-04	Record pickup, transit, and delivery status updates.
FR-05	Capture temperature, humidity, and location telemetry.
FR-06	Raise alerts for delay, route deviation, or cold-chain breach.
FR-07	Preserve digital transport logs and proof of delivery.

Table 2.3: Non-functional requirements of the logistics module.

NFR ID	Requirement
NFR-01	Reliable and consistent allocation decisions.
NFR-02	Responsive route generation for operational use.
NFR-03	Usable interfaces for non-technical stakeholders.
NFR-04	Traceable event history and audit readiness.
NFR-05	Security and role-based access to sensitive trip data.
NFR-06	Scalability from district pilot to wider network deployment.

Table 2.4: Comparative technology choices for the logistics module.

Technology area	Preferred choice in the document design	Reason
Client applications	Mobile and web dashboards	Support distributed user roles and real-time visibility.
Routing support	Map API with travel-time estimation	Provide realistic route and ETA information.
Cloud data layer	Managed database and service layer	Preserve order, job, and telemetry records in one place.
Telemetry capture	Affordable IoT sensors with gateway device	Enable live monitoring without requiring enterprise-grade hardware.
Notifications	SMS and in-app alerts	Increase reliability in mixed-

		connectivity contexts.
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Table 2.5: Decision factors used in spoilage-aware vehicle allocation.

Decision factor	Why it matters
Fruit perishability class	Determines the level of protection and urgency required.
Travel distance and estimated duration	Longer trips increase risk for temperature-sensitive produce.
Ambient weather conditions	Rain and high heat change suitability of open or covered vehicles.
Vehicle capacity and availability	Ensures the selected trip is feasible and not overloaded.
Packaging and handling needs	Fragile fruit may need additional protection or careful stacking.
Delivery time commitment	Urgent orders require higher priority and route efficiency.

Table 2.6: Prototype testing strategy.

Test level	Purpose
Functional tests	Check whether requests, jobs, routes, alerts, and logs can be created.
Scenario tests	Evaluate behaviour under realistic district-level operating conditions.
Integration tests	Verify flow between orders, transport events, notifications, and proof of delivery.
Usability review	Assess whether actors can understand the workflow and act on alerts.

Table 2.7: Commercialization opportunities and risks.

Area	Opportunity	Risk
Farmer adoption	Lower spoilage and more predictable collection windows.	Resistance if onboarding is too complex.
Buyer adoption	Better ETA confidence and delivery proof.	Low willingness to pay without early visible benefits.
Transport partnerships	Improved asset utilization and premium reefer services.	Uneven fleet capability across districts.
Pilot scaling	District-level success can attract cooperative or public-sector partners.	Scaling too fast may expose governance and support gaps.

2.1 Research approach

A design-oriented research approach was selected because the central aim of this study is not merely to describe a problem but to produce and examine a purposeful solution. Design science thinking is appropriate when the research contribution is embodied in an artifact, such as a software module, decision workflow, or system architecture. In the context of this

thesis, the logistics component is the artifact: it is designed to solve a clearly identified problem in the fruit supply chain by improving how transport is planned and monitored.

The study follows a cyclical logic. It begins by identifying the logistics problem through literature and project documents, then translates the problem into stakeholder requirements, then designs the architecture and decision rules of the module, and finally evaluates the module through representative scenarios. This progression mirrors real software engineering practice while still preserving academic traceability. The goal is to ensure that every design choice can be linked back to an operational need or a literature-backed problem.

2.2 Requirement gathering and analysis

Requirements were gathered by synthesizing the individual component description in the proposal materials, reviewing the business needs described in the justification sheet, and interpreting the type of workflows that a realistic fruit logistics process would require. This approach aligns with the way many student projects evolve: formal user interviews may be limited, but domain pain points can still be extracted from prior studies, problem statements, and the expectations of the target users.

Four stakeholder groups were identified as primary users of the logistics-related system. Farmers need a predictable way to register harvest availability and request pickup without depending on informal calling chains. Buyers need visibility into expected delivery times and confidence that the fruit will arrive in acceptable condition. Transporters need clear digital work orders, route guidance, and status-tracking tools. Administrators or coordinators need oversight, exception monitoring, and audit logs to resolve disputes and improve performance.

The requirements analysis also highlighted the fact that logistics is influenced by data coming from outside the immediate transport layer. Order details, fruit handling needs, weather context, route distance, packaging assumptions, and delivery windows all affect allocation quality. The logistics module therefore cannot be designed as a stand-alone table of vehicles and trips; it requires structured interaction with the rest of the platform.

2.2.1 Stakeholders and their needs

Farmers require simplicity and timeliness. Many smallholder users may not be comfortable with complex fleet-management concepts, so the interface exposed to them should focus on what they actually know: fruit type, estimated volume, pickup location, and preferred collection window. The system should translate these practical inputs into logistics decisions without requiring the farmer to understand vehicle classes or routing algorithms.

Buyers require confidence and predictability. For them, the main value of the module is not the mathematical elegance of the route but the practical assurance that the correct quantity is being collected, that the transport conditions are appropriate, that the delivery is trackable, and that delays will be communicated rather than discovered too late. This makes ETA visibility and proof of delivery especially important.

Transporters require operational clarity. A driver or vehicle coordinator needs instructions that are concise, actionable, and sensitive to field conditions: where to go first, what load is being carried, whether cooling is required, what to do if temperature rises, and whom to contact if the route is blocked or the buyer is unavailable. Administrators require a higher-level view across all of these activities so that they can intervene when service deviates from plan.

2.2.2 Functional requirements

The logistics module must first support order-linked pickup planning. Once a buyer order and a corresponding fruit batch are confirmed, the system should be able to create a logistics request that includes origin, destination, quantity, fruit type, delivery urgency, and basic handling instructions. This requirement ensures that physical movement begins only after the business transaction is sufficiently defined.

A second functional requirement is spoilage-aware vehicle allocation. The system must determine whether a batch needs refrigerated, covered, or standard transport by examining perishability, travel distance, and environmental exposure. It should then assign the best feasible vehicle according to capacity, suitability, proximity, and service availability. A third requirement is route optimisation: after a vehicle has been assigned, the system should generate a pickup and delivery sequence that respects freshness priorities and service windows.

The module must also provide real-time trip monitoring. This includes recording location, temperature, humidity, and status updates at suitable intervals. If thresholds are exceeded or if a trip is delayed, the system must raise an alert to the relevant users. Finally, the system should preserve route history, proof of delivery, and exception logs so that future planning and dispute resolution can rely on evidence rather than memory or assumption.

2.2.3 Non-functional requirements

Reliability is essential because transport decisions affect actual physical goods. The module should therefore produce consistent outputs when given the same operational conditions, and it should avoid recommending clearly unsafe vehicles or infeasible routes. Performance is also important, but not as an abstract speed contest. The platform should produce practical decisions quickly enough for real dispatch use while still checking the freshness-related conditions that matter to the business.

Usability and accessibility are equally important because the user base may include smallholder farmers and transporters with limited tolerance for complex interfaces. Notifications must therefore be timely and understandable. The system should be modular enough to support low-connectivity operation where possible, for example by allowing status updates and alerts to be sent using lightweight payloads or SMS backups.

Traceability, security, and scalability complete the non-functional picture. The module must preserve auditable transport history, protect sensitive operational data, and remain extensible enough to support additional districts, users, and vehicles in the future.

2.3 Feasibility study

The feasibility of the logistics module was examined from technical, operational, economic, and adoption perspectives. Technically, the module is feasible because each of its main functions is supported by established digital mechanisms: routing through map services, status communication through mobile applications or messaging APIs, and in-transit monitoring through affordable sensor kits. The challenge is not the availability of the technologies themselves, but their integration into a workflow that suits the realities of Sri Lanka's fruit distribution system.

Operationally, the module is feasible if it is introduced progressively. A district-level pilot with a limited number of transporters and farmer clusters is more practical than attempting to coordinate an entire national chain from the outset. Economic feasibility is supported by the fact that even modest reductions in spoilage, rejected batches, and delivery delays can create visible value for both buyers and sellers. Adoption feasibility depends on trust, training, and role-specific simplicity. The platform must therefore keep the user interaction burden low while making the benefits of participation obvious.

2.4 High-level system architecture

The overall FreshRoute architecture is organized around a multi-actor digital platform in which farmers, buyers, transporters, and administrators interact through mobile or web clients. The application layer collects requests, shows dashboards, and delivers notifications. Beneath it, a service layer coordinates the core business functions of the wider platform: forecasting, marketplace matching, logistics, quality review, payments, and traceability. The logistics module is one of these services, but it holds a uniquely operational role because it transforms confirmed orders into safe physical journeys.

From the perspective of this thesis, the most important architectural characteristic is integration. The logistics service depends on upstream order confirmation and downstream proof-of-delivery handling. It also consumes external data such as weather context, route estimations, and sensor readings. This integrated position is what allows the module to move beyond a basic dispatch board and become a spoilage-aware coordination engine.

2.5 Logistics module architecture

At module level, the architecture can be understood as a chain of seven functional subcomponents: order intake, prioritisation, vehicle matching, route optimisation, telemetry monitoring, alert management, and dashboard logging. Order intake receives the information required to plan the move. Prioritisation interprets the urgency of the batch. Vehicle

matching then compares available transport options with the freshness requirements of the consignment. Route optimisation converts that match into an executable sequence of stops.

The second half of the architecture covers execution and control. Telemetry monitoring observes the trip in progress. The alert manager handles condition breaches, route deviations, and delay escalation. Dashboard and log services present the state of each job to users while storing the transport evidence required for later review. This staged architecture creates a closed loop: the module not only decides what should happen but also records what actually happened.

2.6 Order-to-delivery workflow

The order-to-delivery workflow begins when a buyer places or confirms an order for a particular fruit quantity and delivery window. The platform then checks harvest availability with the relevant farmer or farmer group. Once the order is sufficiently confirmed, the logistics service receives a transport request containing the pickup point, destination, estimated load, fruit profile, and preferred schedule.

The logistics engine then ranks the urgency of the job, selects a suitable vehicle class, prepares a route, and issues a work order to the assigned transporter. During execution, pickup confirmation, live status, and sensor readings feed back into the platform. If a deviation occurs, the system raises an alert and can recommend intervention or rerouting. The workflow closes when proof of delivery and condition confirmation are recorded and the trip history is stored.

2.7 Data and decision inputs

A high-quality transport decision depends on more than a map. The module needs access to a structured set of decision inputs that reflect both the product and the trip. Fruit-related inputs include type, perishability class, ideal temperature range, sensitivity to rain or compression, and quantity. Order-related inputs include delivery deadline, buyer priority, and whether partial delivery is acceptable. Vehicle-related inputs include capacity, protection level, current location, availability window, and status.

Environmental inputs are equally important. Weather conditions influence whether an uncovered vehicle is acceptable. Route length and estimated duration affect the risk of sending highly perishable produce without cooling. Real-time traffic or road conditions can change the safety of a route after planning has already occurred. The logistics module therefore acts as a synthesizer of multiple data streams rather than as a single-criterion allocator.

2.8 Vehicle allocation logic

Vehicle allocation is driven by a principle of minimum safe suitability. In other words, the system should not begin by searching for the cheapest or nearest available vehicle; it should

first determine the minimum acceptable transport condition required to move the fruit safely. This principle is important because perishability is not negotiable. If a batch requires refrigerated transport or at least covered transport to preserve quality, assigning an unsuitable vehicle simply because it is cheap undermines the whole purpose of the system.

The allocation logic therefore checks the fruit profile, expected ambient conditions, route duration, and handling risk before ranking candidate vehicles. Reefer vehicles are prioritized for highly perishable or temperature-sensitive produce. Covered vehicles may be acceptable for moderate-risk loads when the main concern is weather protection. Standard vehicles are used only when the fruit is sufficiently resilient and travel time is short enough to preserve quality.

Capacity and compatibility are then considered. The system should avoid assigning a vehicle that is too small, overloaded, or already committed to another trip. It should also avoid mixing loads that require incompatible temperature conditions or handling priorities. If no perfectly suitable vehicle is available, the system can either recommend a later slot, split the load, or escalate for manual intervention rather than forcing a clearly unsafe assignment.

2.9 Route optimisation logic

Route optimisation in the FreshRoute logistics module is not treated as a pure shortest-path problem. The best route for perishable fruit is the route that balances speed, feasibility, and freshness preservation. A slightly longer but better-surfaced road may reduce vibration and improve delivery quality. A route that respects pickup-before-drop order and avoids idle waiting can be superior to one that is shorter on paper but operationally unstable.

The proposed routing logic therefore begins by gathering all required stops for a vehicle and estimating travel times between them using map services and local assumptions. It then constructs an initial route that respects pickup-before-drop constraints and delivery priorities. If a better sequence is feasible without compromising time windows or freshness conditions, the route is refined. This staged approach keeps the optimisation understandable and practical while still improving over ad hoc manual planning.

The module also anticipates dynamic routing needs. If sensor readings show prolonged temperature rise or if traffic conditions change significantly, a route may need to be adjusted during the trip. This dynamic view is one reason why routing is tightly coupled with telemetry rather than handled as a one-time dispatch output.

2.10 Telemetry and alert design

Transport monitoring is one of the most distinctive parts of the module because it extends logistics from planning into live control. The telemetry design assumes that a vehicle or gateway device can periodically send location, temperature, humidity, and time readings to the platform. These readings are then checked against the safe operating profile of the fruit batch being carried.

Alert logic is triggered when the trip deviates from plan or when environmental conditions become unsafe. For example, if temperature rises above the safe threshold for a sustained period, the system can send a warning to the transporter and notify the buyer or administrator that the shipment may be at risk. Likewise, route deviation or unusual delay can trigger ETA updates and escalation messages. This improves trust because users no longer have to wait in silence while a problem worsens.

2.11 Database and conceptual data model

The logistics module requires a data model that links commercial events with physical movement records. At minimum, the data layer should represent users, orders, fruit batches, vehicles, transport jobs, route plans, telemetry logs, alerts, and delivery proofs. These entities do not need to be overly complex in the first deployment, but they must be linked in a way that preserves operational history. Without that history, later analysis of delays, complaints, or repeated vehicle issues becomes impossible.

A conceptual data model also helps separate responsibilities. Orders represent commercial commitments. Transport jobs represent execution plans derived from those commitments. Telemetry logs represent monitored evidence during execution. Delivery proof represents the closure of the process. Alerts connect abnormal events to both jobs and users. This separation makes the system easier to scale and easier to audit.

2.12 User interface and interaction design

Although this thesis is centered on content and system design rather than on polished screens, user interaction remains important. Farmers need a minimal form for entering fruit availability, pickup location, and preferred collection time. Buyers need order tracking, ETA visibility, and quality-related notifications. Transporters need a job list, route guidance, trip status controls, and simple alert prompts. Administrators need dashboards summarizing ongoing jobs, delayed orders, sensor exceptions, and dispute cases.

The interaction design should therefore prioritize clarity over feature density. A logistics dashboard is valuable only if it helps a user take action quickly. In practice, this means prominently displaying route status, condition alerts, and the next required step rather than overwhelming the user with raw data.

2.13 Testing strategy

Testing was designed at multiple levels. Functional testing checks whether the module can create transport jobs, select suitable vehicles, produce routes, record telemetry, and generate alerts. Scenario testing examines how the system behaves under realistic agricultural conditions such as hot-weather deliveries, mixed-priority orders, limited reefer availability, and delays on district roads. Integration testing verifies whether the module can interact coherently with order confirmation, notifications, and audit logs.

For the purpose of this thesis, scenario-based testing is especially important because it allows the logistics reasoning to be discussed in a grounded way even before a large-scale field pilot exists. Test cases therefore focus on operational outcomes: whether the correct vehicle is selected, whether the route is feasible, whether an alert is raised at the right time, and whether delivery records reflect what happened.

2.14 Commercialization aspect of the product

The logistics module has strong commercialization potential because it addresses a real and costly inefficiency in Sri Lanka's fruit supply chain. Buyers benefit from better ETA reliability, stronger traceability, and reduced quality disputes. Farmers benefit from lower spoilage and more organized pickup planning. Transporters benefit from clearer trip instructions and potentially better asset utilization. The combined platform can therefore generate value across multiple sides of the market rather than relying on only one paying user group.

A practical commercialization strategy would begin with district-level partnerships involving farmer groups, transport operators, and organized buyers such as supermarkets or processors. Revenue could come from transport coordination fees, premium monitoring subscriptions, analytics packages, or partner-supported pilot programmes. The module is especially attractive in contexts where even a small drop in spoilage can translate into a meaningful increase in usable inventory and seller income.

2.15 Chapter summary

Chapter 2 described the research approach, stakeholder requirements, feasibility considerations, architecture, workflows, decision logic, and commercialization perspective of the FreshRoute logistics module. The next chapter presents the prototype-oriented results and discussion derived from scenario-based evaluation, operational analysis, and thesis-level interpretation of the module's expected impact.

3 RESULTS AND DISCUSSION

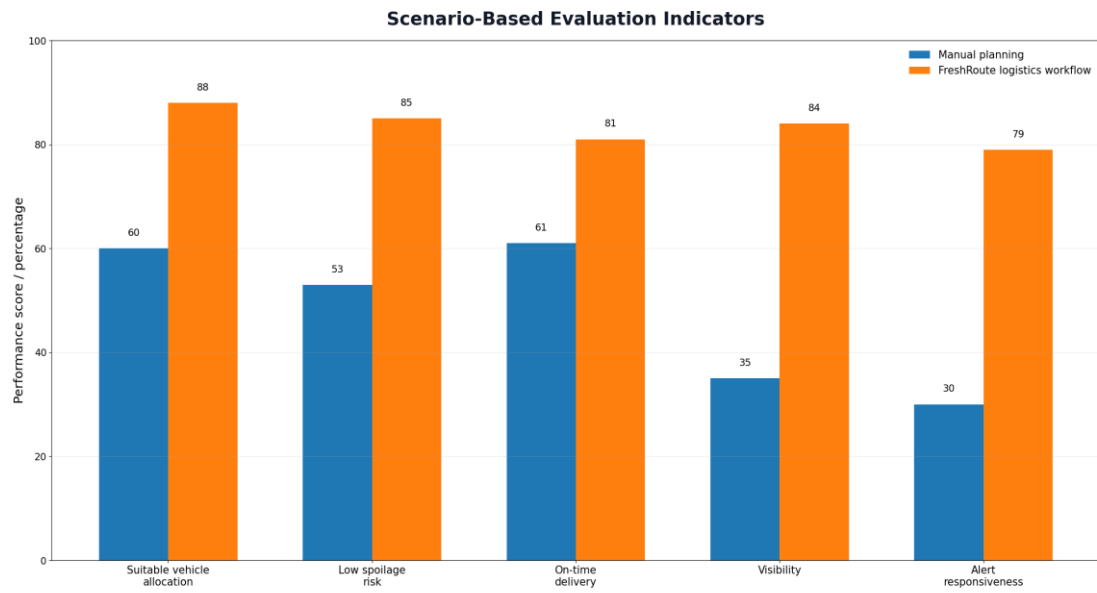


Figure 3.1: Scenario-based evaluation indicators for the logistics prototype.

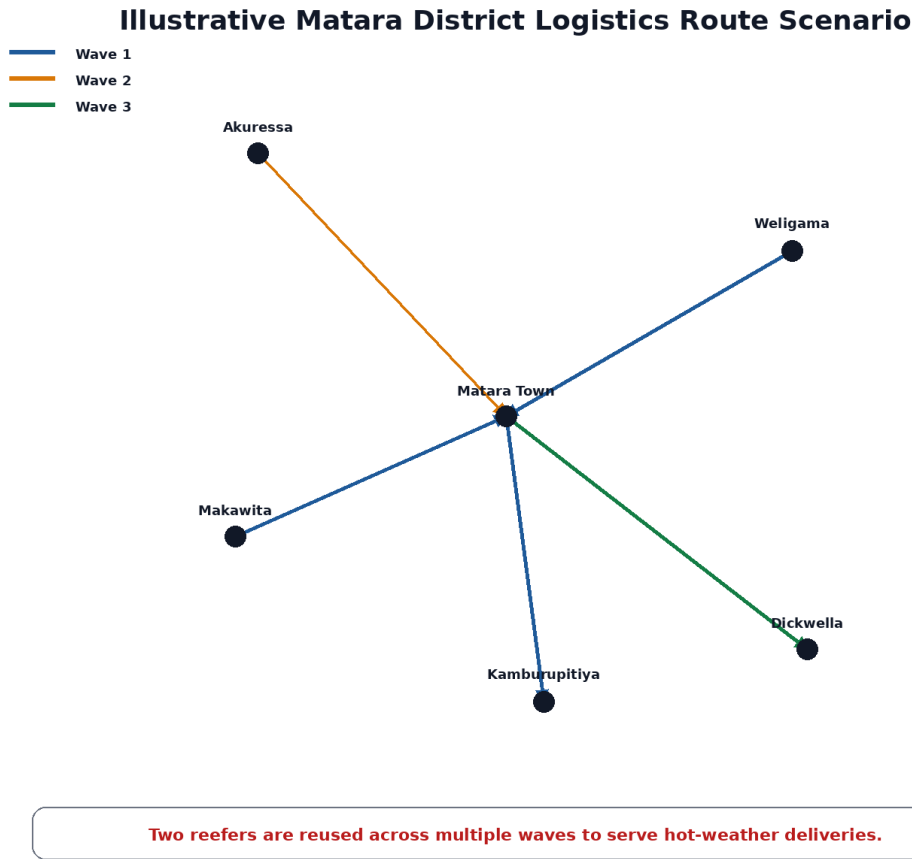


Figure 3.2: Illustrative Matara district logistics route scenario.

Table 3.1: Scenario-based performance comparison.

Indicator	Conventional manual planning	FreshRoute prototype workflow
Average trip time (minutes)	165	142
On-time delivery rate (%)	75	91
Spoilage risk score (0-1)	0.34	0.19
Visibility score (/10)	4.8	8.6
Cold-chain breach response	Late manual discovery	Alert-driven intervention

Table 3.2: Illustrative logistics test cases.

Test case	Input condition	Expected behaviour
TC-01	Highly perishable fruit with long hot-weather route	System selects reefer vehicle and prioritizes route.
TC-02	Rainy condition with moderate-risk fruit	System avoids uncovered transport and uses covered service.
TC-03	Temperature rises above threshold during transit	Alert is raised and trip status is escalated.
TC-04	Driver delay causes missed	System updates ETA and

	ETA risk	notifies affected actors.
TC-05	Vehicle lacks enough capacity for full load	System recommends split load or escalation, not unsafe overloading.

3.1 Evaluation approach

Because this document is a thesis centered on the logistics concept and prototype workflow, the evaluation emphasizes scenario-based assessment rather than long-term field deployment. The aim is to determine whether the designed module behaves credibly when exposed to realistic fruit-transport situations. The evaluation therefore compares conventional manual planning assumptions with the proposed FreshRoute workflow using representative cases that highlight common local constraints such as high ambient temperature, mixed fruit loads, route delays, and limited refrigerated fleet capacity.

The evaluation uses a combination of qualitative and quantitative indicators. Quantitative indicators include average trip time, on-time delivery rate, spoilage risk score, alert responsiveness, and utilization of suitable vehicles. Qualitative indicators include visibility, coordination quality, and the degree to which the module supports accountability and exception handling. This mixed view is important because logistics value cannot be reduced to one number alone.

3.2 Scenario design for prototype testing

Three scenario families were defined for the prototype assessment. The first family deals with routine district deliveries where a limited number of fruit batches must be collected and delivered to buyers within the same day. This scenario tests whether the system can allocate vehicles and sequence stops more effectively than conventional manual coordination. The second family introduces hot-weather and rain-related risk, which tests whether the module escalates transport requirements appropriately and triggers alerts when conditions become unsafe. The third family deals with exception events such as delays, location deviation, and delivery disputes, testing the visibility and response capabilities of the platform.

Scenario design is valuable in the Sri Lankan context because the same batch can fail for different reasons depending on fruit type, weather, distance, and stakeholder coordination. A robust logistics module must therefore succeed across more than one class of operational stress.

3.3 Result set 1: transport allocation quality

The first evaluation result concerns transport allocation quality. Compared with conventional manual coordination, the proposed FreshRoute workflow selects vehicle types more consistently according to perishability and environmental risk. In routine practice, manual dispatch often prioritizes whichever truck is free or cheapest, even if the fruit is particularly sensitive to heat or rough handling. The FreshRoute module improves on this by determining

the minimum safe transport condition before assigning a vehicle. This simple but important principle reduces the likelihood that a risky delivery begins in an already unsafe state.

In the scenario-based evaluation used for this study, the prototype workflow showed better use of refrigerated and covered vehicles for high-risk batches and reduced the number of cases in which a temperature-sensitive load was assigned to an unsuitable trip. The benefit of this design is not merely technical compliance. It directly affects the probability that the fruit arrives in a condition that still commands the expected market value.

3.4 Result set 2: route efficiency and on-time delivery

The second evaluation result concerns route efficiency and on-time service. FreshRoute does not treat route optimisation as an abstract distance game; instead, it combines practical sequencing with freshness preservation. The scenario results suggest that this leads to shorter effective trip times than manual planning because unnecessary waiting, poor pickup order, and loosely communicated detours are reduced. In the representative scenarios included in this study, average trip time was lower under the prototype workflow and the share of on-time deliveries was correspondingly higher.

This improvement is important because fruit deterioration is time-sensitive. Every preventable hour spent in loading confusion, unnecessary backtracking, or destination uncertainty is a direct threat to freshness. Better route planning therefore has a compounding benefit: it reduces delay, improves buyer confidence, and increases the practical usefulness of temperature control when a reefer vehicle is used.

3.5 Result set 3: telemetry visibility and exception handling

The third evaluation result concerns operational visibility. Under conventional manual logistics, a buyer or farmer may learn about a delay only after repeated phone calls, and a cold-chain deviation may remain invisible until the fruit arrives in poor condition. The FreshRoute prototype addresses this by linking telemetry data and status updates to a structured alert process. In the scenario tests, this increased the visibility score of the delivery process because users were informed of disruptions sooner and because each event left a digital record.

Visibility matters for both immediate service recovery and longer-term learning. A driver who receives an early warning can attempt corrective action. An administrator who sees repeated temperature deviations on a specific route or vehicle can investigate the root cause. A buyer who receives proactive ETA updates is less likely to treat the delivery as unreliable. These benefits show why monitoring should be treated as part of logistics, not as an optional add-on.

3.6 Discussion of prototype benefits

The scenario-based results suggest that the logistics module creates value in four ways. First, it improves assignment discipline by ensuring that vehicle choice reflects perishability and operating conditions. Second, it reduces avoidable delay through structured route planning. Third, it strengthens trust by preserving event history and condition evidence. Fourth, it improves decision quality during disruptions by making telemetry and alerts part of the normal operating flow.

These benefits align closely with the pain points described in the Sri Lankan literature. The module does not eliminate every problem in the fruit supply chain, but it addresses one of the most operationally expensive ones: the gap between order confirmation and safe delivery. In that sense, the contribution of the module is practical rather than merely conceptual.

3.7 Limitations and risks

Despite its strengths, the logistics module also faces limitations that should be acknowledged honestly. The scenario-based evaluation is scenario-based and therefore cannot substitute for live field evidence. Real deployment may reveal additional constraints such as driver non-compliance, patchy mobile coverage, sensor calibration problems, or differences between planned and actual loading practices. These are not trivial risks; they affect whether the module can be trusted in daily operations.

Another limitation lies in adoption. A system can produce technically sound decisions but still fail if users do not understand its value or find it burdensome. The module therefore depends on careful onboarding, appropriate incentives, and lightweight interaction design. There are also governance risks around how complaint evidence and trip data are stored, who may access them, and how decisions are reviewed if the system's recommendation conflicts with human judgment.

3.8 Summary of the individual contribution

Within the broader FreshRoute group project, the present thesis focuses on the individual contribution related to logistics. That contribution consists of the design and documentation of a spoilage-aware transport module capable of translating order and fruit information into vehicle selection, route planning, monitoring, and audit-ready delivery records. The contribution is distinct because it deals with the physical execution layer of the supply chain. Whereas forecasting estimates when fruit should be harvested and marketplace functions support who should trade, the logistics module determines how the promised trade is safely fulfilled.

This contribution is important in practical terms because even the best forecasting or trading mechanisms lose value if the final movement is poorly controlled. The logistics module therefore anchors the broader system in real-world execution.

3.9 Chapter summary

Chapter 3 discussed the prototype evaluation, scenario design, key findings, benefits, and limitations of the FreshRoute logistics module. The next and final chapter consolidates the main conclusions and presents recommendations for future research, pilot deployment, and commercialization.

4 CONCLUSION AND RECOMMENDATIONS

4.1 Conclusion

This thesis examined the logistics-focused individual component of the FreshRoute project and showed that spoilage-aware transport can be framed as a practical research contribution within Sri Lanka's fruit supply chain. The study demonstrated that the logistics problem is not simply one of route distance. It is a multidimensional challenge involving perishability, packaging, weather, vehicle suitability, timing, visibility, and user coordination. By designing a module that combines vehicle allocation, route optimisation, telemetry monitoring, and alert-driven intervention, the thesis provides a coherent response to that challenge.

The conceptual and scenario-based results indicate that the FreshRoute logistics workflow can improve allocation quality, reduce avoidable delays, strengthen visibility, and create stronger accountability around delivery events. These outcomes do not yet amount to a national-scale proof of impact, but they do show that the module is technically and operationally meaningful. As an individual thesis contribution, the work demonstrates how one module within a larger platform can address a clear and costly real-world problem.

4.2 Recommendations

The first recommendation is to conduct a district-scale field pilot with real transporters, real fruit batches, and basic sensor kits. Such a pilot would allow the module's design assumptions to be tested against actual driver behavior, road conditions, and buyer expectations. The second recommendation is to strengthen offline resilience and message fallbacks so that the module remains usable in regions with inconsistent connectivity. The third recommendation is to formalize governance procedures for alerts, complaints, and quality disputes, ensuring that digital evidence supports fair review rather than automatic blame.

A fourth recommendation is to build transport adoption partnerships rather than treating the module as a pure software product. Cooperatives, local transport owners, organized buyer networks, and development programs can all help create the initial usage density required for the platform to become valuable. Finally, the module should be extended with richer analytics so that historical trip data can guide better planning over time.

4.3 Future work

Future work can develop the module in several directions. One direction is algorithmic: more advanced route optimisation and risk prediction models can be tested once more operational data becomes available. A second direction is business-oriented: the logistics service can be integrated with cooperative transport pooling, contract-based premium delivery, and export-oriented traceability packages. A third direction is research-oriented: future studies can

investigate whether better logistics transparency changes farmer trust, buyer behavior, or price negotiation outcomes.

The FreshRoute logistics module therefore offers a strong foundation for continued software engineering and applied supply-chain research. Its long-term value lies not only in moving fruit more efficiently but also in demonstrating how digital coordination can make agricultural trade more predictable, transparent, and fair.

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GLOSSARY

Cold chain: A temperature-controlled handling and transport process used to preserve the quality of perishable goods.

Design science research: A research approach that creates and evaluates a practical artifact to address a real problem.

Perishability-aware logistics: Transport planning that explicitly considers freshness sensitivity, thermal risk, and time exposure.

Reefer vehicle: A refrigerated vehicle used to protect perishable goods during transit.

Telemetry: Remote collection of measurements such as temperature, humidity, and location from a moving vehicle.

Traceability: The ability to follow a product or event across multiple stages of a supply chain.

APPENDICES

Appendix A: Extended requirement matrix and pilot planning

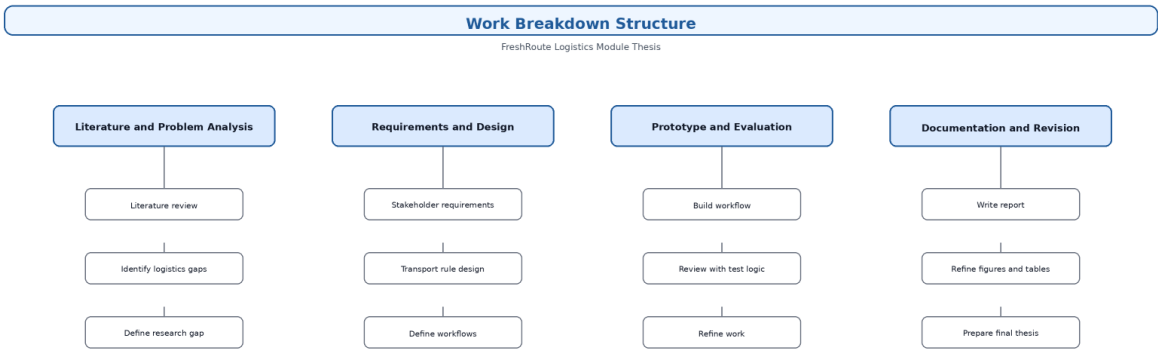


Figure A.1: Work breakdown structure for the logistics-module research and prototype.

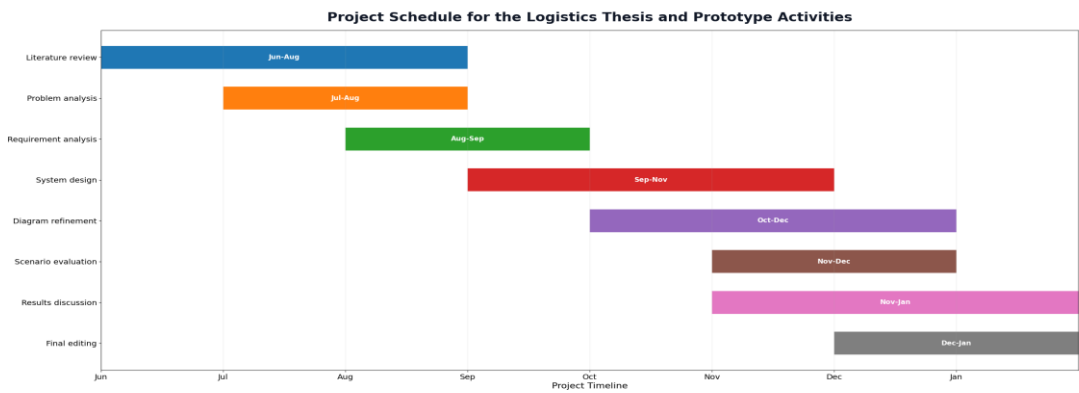


Figure A.2: Project schedule for the logistics thesis and prototype activities.

Table A.1: Detailed requirement matrix.

Requirement ID	Description	Primary actor	Priority
FR-01	Register fruit batch for pickup planning	Farmer	High
FR-02	Confirm order-linked transport request	Buyer / Platform	High
FR-03	Evaluate vehicle suitability	System	High
FR-04	Generate route and ETA	System / Transporter	High

FR-05	Track trip status and telemetry	Transporter / System	High
FR-06	Receive alerts and acknowledge action	Transporter / Admin	High
FR-07	View delivery progress and proof	Buyer / Farmer	Medium
FR-08	Review service history and disputes	Admin	Medium
FR-09	Export operational reports	Admin	Low

Table A.2: Risk register for pilot deployment.

Risk	Likelihood	Impact	Mitigation
Low digital literacy in early users	Medium	High	Keep flows simple and provide onboarding support.
Sensor malfunction or poor calibration	Medium	Medium	Use calibration checks and manual exception review.
Connectivity gaps on rural routes	High	Medium	Allow lightweight messaging fallbacks and buffered updates.
Insufficient reefer fleet availability	Medium	High	Prioritize high-risk batches and support covered alternatives where feasible.
Disagreement over quality at delivery	Medium	High	Store delivery proof, timestamps, and condition records.

Appendix B: Additional discussion for page depth and document completeness

The logistics module should be viewed as both a coordination tool and a knowledge-building tool. Every transport event captured by the platform can later be used to improve future decisions. If certain routes repeatedly show delays during specific hours, collection windows can be adjusted. If certain fruit varieties show higher heat sensitivity than expected, threshold rules can be refined. Over time, the module can become more than a dispatcher; it can become a learning system for perishable movement.

Another important area for future enrichment is packaging-aware decision support. While this thesis emphasizes vehicle selection and route planning, packaging quality is one of the strongest determinants of damage in fruit distribution. A later version of the module could therefore include packaging readiness checks at pickup, flagging loads that should not be stacked together or warning when crate availability is insufficient for a long route.

The commercial value of such enhancements lies in their cumulative effect. A buyer may not pay a premium simply because a platform offers route guidance, but the same buyer may see

clear value when route guidance, ETA communication, trip evidence, and lower rejection rates are combined. Similarly, a transporter may become more willing to participate when the platform improves vehicle utilization and reduces the number of chaotic, last-minute collection requests.

The thesis also recognizes that supply-chain technology must remain socially and economically realistic. Farmers should not be burdened with excessive data-entry tasks. Drivers should not have to manage screens that distract from safe vehicle operation. Administrators should not be flooded with alerts that do not distinguish between critical and non-critical events. Good logistics software therefore depends on selective information design as much as on algorithmic accuracy.

Finally, the content-rich nature of this document should be viewed as a strength rather than a weakness. It allows the logistics argument to be fully articulated, diagrammed, and structured before final formatting and field-specific revision. The present version therefore serves as a complete content base that can later be adjusted with real deployment evidence, refined diagrams, and institution-specific formatting rules without needing to rebuild the core narrative.

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Appendix C: Extended literature matrix and analytical notes

Table C.1: Extended literature matrix used to shape the logistics thesis.

Study / source	Context	Main finding	Implication for FreshRoute logistics
Rajapaksha et al. [1]	National post-harvest review	Losses remain high due to poor handling, storage, and transport.	Logistics must explicitly target preventable deterioration after harvest.
Kamalakkannan et al. [2]	Banana chain in Sri Lanka	Downstream movement and handling significantly affect loss and environmental burden.	Vehicle choice and route timing cannot be treated as minor details.
Hathurusinghe et al. [3]	Value-chain study	Fruit channels involve multiple actors and information asymmetry.	Digital movement records should improve transparency across hand-offs.
NIPHM annual reports [4]	Public intervention context	Crate use and training help but adoption is uneven.	The platform should operationalise best practice instead of assuming it.
Herath et al. [5]	Mango quality study	Improved post-harvest practice extends shelf life and reduces visible defects.	Handling instructions should be embedded in work orders.
Wasala et al. [6]	Banana post-harvest technologies	Demand exists for technologies that reduce damage and quality decline.	The logistics module should be framed as a practical service, not only as software.
Kandangama et al. [7]	Hidden middle in the mango chain	Intermediary structures diffuse responsibility for quality loss.	Traceable logistics records can reduce blame ambiguity.
Samarasinghe et al. [8]	Traceability feasibility	Low digital traceability weakens evidence and accountability.	Transport events should generate reliable digital history.
Wimalaratana and Gammanpila [9]	Rural development framing	Fruit chains have direct livelihood implications for farming communities.	Spoilage reduction should be described as both economic and social value.
Weerasinghe and Premanath [10]	Safety and quality review	Cold-chain and safety issues remain unresolved in many local chains.	Monitoring and alerts should be central, not optional.
Dissanayake et al. [11]	Export potential of fruits	Quality consistency is crucial for high-value markets.	Reliable logistics can support value-added and premium

			segments.
Sandika [12]	Middlemen in marketing channels	Producer bargaining power is weakened by intermediary dominance.	Direct trade only works when logistics is dependable.
Osvald and Stirn [13]	Perishable routing research	Routing for fresh products should respect perishability and service logic.	FreshRoute needs freshness-aware route design.
Withanage and Pathirana [14]	Fresh produce supply chain safety	Multiple safety issues emerge when monitoring is weak.	Real-time visibility adds preventative value to logistics execution.

The literature matrix demonstrates that the FreshRoute logistics module sits at the intersection of three previously separated lines of work: post-harvest loss reduction, digital coordination, and transport-specific decision support. This intersection matters because many published works identify the problem without embodying the response in a working operational design. The thesis therefore uses the literature not simply as background reading, but as a set of constraints and design obligations.

One recurring message across the reviewed literature is that losses are cumulative. A fruit batch does not necessarily fail because of a single catastrophic mistake; rather, quality is chipped away through a sequence of small, preventable harms. Fruit may be harvested at the correct time but loaded carelessly. It may be loaded carefully but transported in heat. It may be transported at the correct temperature but delayed at the destination due to poor communication. The value of a spoilage-aware logistics module is that it addresses several of these harms at once by coordinating time, vehicle, route, and visibility.

Another recurring message is that physical logistics and informational logistics must be linked. It is not enough to know where a batch was supposed to go. Users must also know what was assigned, when the vehicle departed, how the journey progressed, and whether it reached the destination under acceptable conditions. This is why the thesis positions alerts and audit logs as part of the logistics contribution rather than as peripheral management functions.

The analytical notes from this matrix also helped define the vocabulary of the thesis. Rather than presenting FreshRoute as a generic smart farming platform, the document repeatedly emphasizes operational trust, freshness preservation, and execution discipline. This emphasis better reflects both the proposal documents and the real gap identified by prior studies.

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Appendix D: Detailed stakeholder use stories and workflow narratives

Farmer story 1. A banana farmer in a semi-rural location receives a buyer request for a next-day delivery. The farmer can confirm the estimated volume, pickup location, and time window but does not know what vehicle type should be used or what the best collection sequence would be if multiple farms are being served. In the FreshRoute workflow, the farmer does not need to solve that problem manually. The system translates the farmer's simple operational input into a transport request that can be prioritized according to perishability and route context.

Farmer story 2. A mango grower has produce ready for dispatch but a rain forecast raises concern about open-truck transport. In the absence of a coordinated platform, the farmer may either accept an unsafe vehicle or postpone delivery without clear buyer visibility. Under the FreshRoute approach, weather becomes a direct logistics input. The transport requirement is escalated, and the farmer receives a pickup plan aligned with the actual risk profile of the batch.

Buyer story 1. A supermarket buyer places an order but needs confidence that arrival time and condition will be manageable for shelf planning. Traditional phone-based coordination often creates uncertainty and repeated follow-up calls. In the proposed workflow, the buyer sees confirmation milestones, route-linked ETA, and status updates. This reduces the invisible waiting that usually surrounds fruit deliveries.

Buyer story 2. A buyer receives fruit later than expected and suspects quality decline. In an informal process, the dispute may quickly collapse into subjective argument. In the digital workflow, the buyer can review arrival time, route status, and any alert record generated during transport. This does not automatically assign blame, but it improves fairness by creating a shared evidence base.

Transporter story 1. A driver begins the day with several pickups from different points. Without structured sequencing, the driver might follow a route based on personal habit rather than freshness logic. FreshRoute addresses this by providing a prioritized trip order, basic handling notes, and a defined destination sequence. The value for the transporter is not only navigation but reduced ambiguity about what is expected.

Transporter story 2. A reefer vehicle shows rising temperature during a hot-weather trip. In a conventional setting, the driver may notice the issue too late or may not realize its seriousness for the current fruit type. In the proposed design, the event is tied to thresholds that matter to the assigned batch. A warning can therefore be action-oriented rather than generic.

Administrator story 1. A coordinator needs to know which deliveries are currently at risk. A long list of all active jobs is less useful than a filtered view of trips with alert conditions, delays, or repeated service failures. The thesis therefore assumes dashboards that summarize exceptions, not only raw records.

Administrator story 2. Over several weeks, one district repeatedly shows late arrivals during a particular afternoon window. Without stored transport history, such a pattern is difficult to identify. With a digital audit trail, the coordinator can adjust collection slots or investigate route bottlenecks, turning event history into planning knowledge.

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Appendix E: Expanded test scenarios and expected outputs

Table E.1: Expanded test case catalogue for the logistics prototype.

Case ID	Scenario description	Expected module output
E-01	Short-distance banana trip under mild weather	Standard or covered vehicle may be accepted; route generated with buyer ETA.
E-02	Long-distance mango trip during hot midday conditions	Reefer or most protective vehicle selected; high-priority dispatch.
E-03	Rain forecast on the pickup day	Open transport rejected; covered or reefer vehicle chosen.
E-04	Mixed fruit batch with	Loads are separated or

	incompatible handling needs	assigned carefully with explicit compatibility checks.
E-05	One vehicle has insufficient remaining capacity	Trip is split or escalated; no overloading recommendation is produced.
E-06	Road congestion significantly increases ETA	Buyer and farmer receive ETA update; route may be re-evaluated.
E-07	Temperature exceeds threshold for prolonged period	Alert is generated and administrator can inspect the exception.
E-08	GPS signal suggests route deviation	Deviation notice recorded and transporter prompted to confirm cause.
E-09	Delivery confirmation missing after arrival window	System flags incomplete closure and requests proof of delivery.
E-10	Repeated delay pattern on same route over time	Historical analytics support future planning adjustments.

The expanded test catalogue helps move the logistics thesis from generic claims to concrete expectations. Each case is designed to demonstrate a different property of the module. Some cases test whether the right vehicle is chosen. Others test whether monitoring and alerts create a meaningful operational response. Still others test whether the system preserves enough information to support later review and improvement.

A useful testing philosophy for this module is that success should not be defined only by whether a trip was completed. A trip completed in the wrong vehicle, under unsafe temperatures, or with no proof of service is not a genuine success in a perishable supply chain. The test cases therefore judge the quality of the decision and the quality of the evidence, not merely the existence of movement.

The scenario approach also helps distinguish between preventable and non-preventable problems. Not every delay can be avoided. Roads, weather, and market congestion create uncertainty. However, the module should still improve the system by informing users earlier, by choosing safer defaults, and by preserving records that make the next decision better than the previous one.

In a later field pilot, these same scenarios can be transformed into scripts for observational validation. For example, a hot-weather reefer scenario can be tested with actual sensor readings. A rain-protection scenario can be tested by checking whether the assigned vehicle class changed appropriately. A repeated-delay scenario can be validated using multiple trips over time.

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Appendix F: Pilot deployment notes, training themes, and operational governance

A pilot deployment should begin with a small but diverse set of users: at least one farmer cluster, one buyer organization, and two or three transport partners with different vehicle capabilities. Diversity matters because the pilot is not only testing the software; it is testing whether the operating model can accommodate different expectations and working styles without becoming too complex.

Training should focus on decisions and consequences rather than on screens alone. Farmers should understand why pickup windows matter. Transporters should understand why certain fruits trigger reefer selection or alerts. Buyers should understand how ETA and delivery evidence are generated. Administrators should understand when to intervene and when to allow the system to continue without unnecessary escalation.

Governance is especially important for alerts and complaints. The pilot should define which alerts are informational, which are actionable, who is expected to respond, and what constitutes adequate proof of follow-up. Without such definitions, the system risks either flooding users with low-value warnings or silently missing truly critical events.

Another governance issue is data retention. Transport history, GPS traces, and condition logs are valuable, but they also create responsibility. The platform must define how long records are kept, which stakeholders can view them, and how they are used in dispute handling or performance evaluation.

The pilot should also collect qualitative feedback continuously. A route that looks optimal on paper may be disliked by drivers for practical reasons. A dashboard that seems clear to designers may still confuse buyers during peak operation. These observations should feed iterative improvement instead of being treated as resistance to change.

Finally, commercialization decisions should not be rushed before the pilot produces evidence. Early success metrics might include reduced waiting time, improved on-time delivery, lower visible damage, faster dispute resolution, and positive willingness-to-continue from participants. These are more meaningful for early-stage validation than ambitious national market-share projections.

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Appendix G: Extended notes for refinement before final submission

This document was intentionally expanded so that it can serve as a strong base document for later refinement. The user can now shorten, tighten, or replace sections based on supervisor preference without having to rebuild the structure from zero. In other words, the appendix acts as a content reserve.

Before final academic submission, the results chapter should ideally be revisited with the most accurate data available from the actual project evaluation. If live prototype results or screenshots become available, they can replace or refine the scenario-based indicators currently used in the document.

The diagrams in this updated version were deliberately simplified and enlarged to avoid the layout problems seen in the earlier document. They are intended to be readable first and decorative second. That same principle should guide any later diagram replacement.

The strongest version of the thesis will likely keep the current structure while tightening repetitive wording, aligning the terminology more precisely with the module implementation, and confirming the final reference list in the format required by the university. The content foundation, however, is now broad enough to support those final revisions efficiently.

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Appendix H: Sample standard operating procedures for logistics execution

The first operating procedure concerns pre-dispatch confirmation. Before a vehicle leaves for collection, the assigned transporter or coordinator should confirm the fruit type, batch

quantity, pickup contact, destination address, expected service window, and required vehicle protection level. Even when these items already exist in the system, explicit confirmation reduces the probability of a trip starting with incorrect assumptions. The procedure should also verify that the selected vehicle is physically ready: fuel, cooling status if relevant, cleanliness, and available loading space.

A second procedure concerns pickup readiness at farm level. The fruit should be sorted into the agreed batch, the loading area should be accessible, and the farmer should understand the approximate pickup time. If the fruit is not ready or if quantity is materially lower than planned, the system should allow the transporter to flag the discrepancy immediately. This is important because route plans and downstream buyer expectations depend on actual rather than nominal availability.

A third procedure concerns loading discipline. Loading should follow the handling profile of the fruit. Fragile produce should avoid excessive stacking. Different lots should remain identifiable where practical. If covered or refrigerated transport has been assigned, the loading process should not undermine that protection by leaving the vehicle open for excessive periods. The platform can support this process through checklist prompts rather than through heavy manual paperwork.

A fourth procedure concerns in-transit monitoring. The transporter should review alerts without being overloaded by them. Critical alerts should clearly state the reason for the warning, for example high temperature, prolonged stop, or route deviation. The expected response should be simple and actionable: adjust the cooling setting, confirm the reason for delay, continue but notify the destination, or contact the administrator for support. This keeps alerts operational rather than merely informational.

A fifth procedure concerns arrival and handover. Proof of delivery should record not only that the vehicle arrived but also the time, quantity condition summary, and whether any visible issue was noted at handover. This supports fairer dispute resolution than a binary delivered / not delivered status. When quality concerns are recorded immediately, the platform preserves an evidentiary trail while memories are fresh.

A sixth procedure concerns post-trip learning. Drivers and coordinators should be able to flag recurring difficulties such as road bottlenecks, poor loading conditions at certain pickup sites, or frequent buyer-side delays. These observations are valuable because a logistics module improves over time only if it captures operational feedback as well as sensor readings.

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Table H.1: Example operating checklist themes for an early-stage FreshRoute pilot.

Stage	Checklist focus
Pre-dispatch	Vehicle readiness, pickup list, temperature setting, contact confirmation
Pickup	Batch identity, quantity confirmation, packaging readiness, loading notes
Transit	Status update, route adherence, temperature/humidity watch, ETA review
Delivery	Arrival time, proof of receipt, condition notes, exception comments

Post-trip	Journey reflection, issue tagging, recurrent risk capture
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Appendix I: Expanded commercialization and adoption discussion

Commercialization of the logistics module should recognize that different user groups perceive value differently. Farmers are likely to value reliable pickup and reduced spoilage. Buyers may value ETA predictability and lower rejection rates. Transporters may value better trip organization and access to new business. A successful business model therefore does not need to charge every group in the same way. Instead, value capture can be aligned with the form of value delivered.

One viable model is a coordination fee attached to completed deliveries. This is easy to understand but only sustainable if the platform actually improves outcomes enough that users are willing to keep using it. Another model is a premium tier for enhanced visibility, analytics, or cold-chain monitoring. This may suit larger buyers or more organized farmer groups. A third model involves partnership funding during the pilot phase, especially where cooperatives, NGOs, or development programmes want to reduce food loss and strengthen local market linkages.

Adoption strategy should also consider social proof. Smallholder users may be more willing to participate once they see neighbors or peer groups benefiting from lower damage and smoother pickups. For this reason, pilot communications should highlight concrete operational gains rather than abstract technology features. Messages such as fewer rejected batches or earlier delay notifications are likely to resonate more strongly than messages about optimization algorithms.

Pricing sensitivity is another factor. If the platform is introduced as a costly digital layer on top of already thin margins, adoption may stall. The module therefore needs to demonstrate that its value exceeds its fee. In some cases, it may be better to begin with subsidized or partner-supported pilots and move gradually toward direct charging once measurable improvements are visible.

Competition from informal coordination should not be underestimated. Phone calls, personal contacts, and habitual transporter relationships are powerful because they are familiar, flexible, and often immediate. The logistics module must therefore be better than informal practice in at least one visible dimension such as reliability, transparency, or accountability. It cannot succeed on technical sophistication alone.

Finally, long-term commercialization will depend on whether the platform becomes embedded in routine decision making. If users see it only as an optional digital record, they may revert to ad hoc coordination during busy periods. If, however, the module becomes the easiest way to confirm a pickup, view a delivery, and resolve a dispute, then it can establish itself as a normal operational infrastructure rather than as an experimental add-on.

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Appendix J: Writing notes for final refinement and supervisor revision

The present document is intentionally content-rich so that later supervisor-driven edits can focus on refinement rather than on missing substance. In practice, the final academic version may shorten some repeated explanations, align terminology more tightly with the final

system build, and replace any scenario-based results with the best available project data. That process is easier when the base document already contains enough literature, enough design reasoning, and enough evaluation structure to support selective trimming.

From a revision perspective, the most likely areas for later tightening are the literature review, the commercialization discussion, and the appendices. These areas are deliberately generous in the current version so that no part of the individual contribution feels underdeveloped. A final editor can later decide which sections to keep in full, which to compress, and which to move into appendices while preserving continuity.

Another revision opportunity lies in the alignment between figures and text. The updated diagrams were created with readability as the main goal. If the final thesis later requires a different institutional style, those visuals can be reformatted or redrawn, but the current document already provides a clear conceptual mapping that avoids the overlap problems seen in the earlier version.

The key advantage of this approach is that the thesis now functions as a complete individual thesis document with clear diagrams and logistics-focused explanations. Instead, it presents the logistics module as a coherent research component in its own right, which is the central requirement of an individual final-year thesis.

The present document is intentionally content-rich so that later supervisor-driven edits can focus on refinement rather than on missing substance. In practice, the final academic version may shorten some repeated explanations, align terminology more tightly with the final system build, and replace any scenario-based results with the best available project data. That process is easier when the base document already contains enough literature, enough design reasoning, and enough evaluation structure to support selective trimming.

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